E-K1: a Gauss-period window in s = x + 1/x on Koblitz analogues (run-20261019T1712Z)

Kind: solver-axis stage diagnostic (AGENTS.md §2). Observations only. Nothing here prices a pipeline, reports S or shows a speedup. The protocol is `PREREGISTRATION.md` Amendment 6 (with its erratum), registered before any row. Amendment 6 follows the rule of Amendment 5 (κ is the primary score): $\kappa = \Delta\log_2$ field_ops / Δb against the break-even m − 1.

Run

- **Commands** (`B=fb_koblitz_ek1`, `R=research/factor_base_designs_20261015/runs`, one series at a time under `flock /tmp/claude-0/measure.lock`, each series followed by its replay):
  `$B check --n 11 --ell-max 8; $B check --n 23 --ell-max 8` (log `runs/K1-check.txt`)
  `$B run --arms A --n N --m M --out-dir $R` for A in S, X, XR, U, NS (and PC at M = 3), N in 11, 23, M in 2, 3
  `$B replay $R --out K1-replay.jsonl --arm A --n N --m M` after each series
  `$B analyze $R --replay K1-replay.jsonl > $R/analysis-amendment6.txt`
- **Instrument:** `pq_f4_f2::groebner_basis_f4` (the repository's Boolean F4 over $F_2[v]/(v^2 - v)$, Gebauer–Möller pairs and field pairs) on the Weil restriction of every system. D = `solving_degree`, field_ops = `word_xors` (64-bit XORs of the row reductions), max_cols = `matrix_cols_max`, max_rows = `matrix_rows_max`. The same engine, 300 s budget, 420 s hard kill and `ulimit -v 13000000` for every arm.
- **Binary:** built at `b733807e0` (every row records it). The analysis marks for P6–P8 were added afterwards (`6e1b329ce`); no rule changed.
- **Host:** "vm", Intel Xeon @ 2.80 GHz, 4 cores, 15 GB RAM, a Claude Code cloud container shared with other agents, serialised by the lock; no further isolation. **Window:** 2026-10-19, 17:12Z to 20:28Z.
- **Field and curve:** K_0 over F_2^n, n = 11 and 23, `find_irreducible_sparse(n)` modulus; ζ of order 2n + 1 = 23, 47; Gauss periods $c_i = \zeta^i + \zeta^{-i}$. `KoblitzCurve::new` rejects K_0 at n = 11 (`#E = 4·23²`), so the curve is built directly; `#E` = 2116 and 8 383 412 was confirmed by counting every abscissa.
- **Records:** `runs/K1-{S,X,XR,U,NS}-n{11,23}-m{2,3}.jsonl`, `runs/K1-PC-n{11,23}-m3.jsonl`, each with its `.log` `runs/K1-replay.jsonl` `runs/K1-check.txt` `runs/analysis-amendment6.txt`. 306 rows, 22 of them `no_plant` placeholders (see below).
- **Seeds:** `0x454B31 ⊕ n≪48 ⊕ arm≪40 ⊕ m≪32 ⊕ ℓ≪16` target k from `cell ⊕ (k+1)≪8`. S, X, XR, U, NS, PC = arm 0–5.

Exact facts checked before any measurement (class: exact computation)

- **Identity.** Eliminating σ from the repository's symmetrised polynomial (w = u² + u, s = 1/w) gives a polynomial G with $G(1/s)\cdot(\Pi\, s)^k$ = Semaev's S_{m+1} of `plain_terms(m)` (b = 1), on a 6-per-variable grid (degree ≤ 4 per variable, so exact) at n = 11 and 23, m = 2 and 3. So **S and X have identical equations**; they differ in the target value (s_R against x_R) and in which curve points the window contains.
- Every set (independent basis, exact point count against `points_with_x` where cheap), every planted point (on the curve, coordinate in the window, sum = R), the word-field value of every polynomial at the planted point and the Boolean equations at the planted bit-vector, were checked inside each system's build. No check failed.

Series

Cells are completed when both targets are ok. κ and σ are least-squares slopes of $\log_2$ cell field_ops and of cell D on b = $\log_2$ of the exact factor-base point count. "Limit" is the first cell where a system hit the budget (timeout) or a Macaulay matrix would exceed 1 GiB (oversize); the series stops there.

arm	n	m	cells (ℓ)	limit cell	κ	σ	κ verdict (m − 1)
S	11	2	9 (3–11)	none (ℓ = n)	3.932	0.403	unfavourable (1)
X	11	2	9 (3–11)	none	3.581	0.384	unfavourable (1)
XR	11	2	9 (3–11)	none	3.783	0.397	unfavourable (1)
U	11	2	9 (3–11)	none	3.642	0.352	unfavourable (1)
NS	11	2	9 (3–11)	none	3.849	0.438	unfavourable (1)
S	23	2	11 (3–13)	ℓ = 14 oversize	2.791	0.226	unfavourable (1)
X	23	2	11 (3–13)	ℓ = 14 oversize	2.424	0.199	unfavourable (1)
XR	23	2	11 (3–13)	ℓ = 14 oversize	2.849	0.247	unfavourable (1)
U	23	2	10 (4–13)	ℓ = 14 oversize	3.077	0.279	unfavourable (1)
NS	23	2	11 (3–13)	ℓ = 14 oversize	2.750	0.467	unfavourable (1)
S	11	3	2 (4–5)	ℓ = 6 timeout	7.331	-	inconclusive (2)
X	11	3	2 (4–5)	ℓ = 6 timeout	6.957	-	inconclusive (2)

XR	11	3	3 (3–5)	ℓ = 6 timeout	9.397	0.000	unfavourable (2)
U	11	3	3 (4–6)	ℓ = 7 oversize	10.036	0.770	unfavourable (2)
NS	11	3	2 (4–5)	ℓ = 6 timeout	7.072	-	inconclusive (2)
PC	11	3	2 (4–5)	ℓ = 6 oversize	3.694	-	inconclusive (2)
S	23	3	2 (4–5)	ℓ = 6 oversize	5.237	-	inconclusive (2)
X	23	3	2 (4–5)	ℓ = 6 oversize	3.778	-	inconclusive (2)
XR	23	3	3 (3–5)	ℓ = 6 timeout	18.707	1.827	unfavourable (2)
U	23	3	3 (4–6)	ℓ = 7 oversize	6.148	0.000	unfavourable (2)
NS	23	3	2 (4–5)	ℓ = 6 timeout	5.007	-	inconclusive (2)
PC	23	3	0	ℓ = 4 oversize	-	-	inconclusive (2)

σ is shown only where at least 3 cells completed. `no_plant` rows: at m = 3, ℓ = 3 holds no planted decomposition with 3 distinct summands for S, X, U, NS and PC at both n (XR's random subspace did hold one), and U at n = 23, m = 2, ℓ = 3 holds none; those cells are skipped, not limits. XR and U draw an independent random window for every ℓ, so their b is not monotone in ℓ: XR at n = 23, m = 3 has |F| = 19 at ℓ = 4 and at ℓ = 5, which makes its κ (18.707) a statement about b, not about cost.

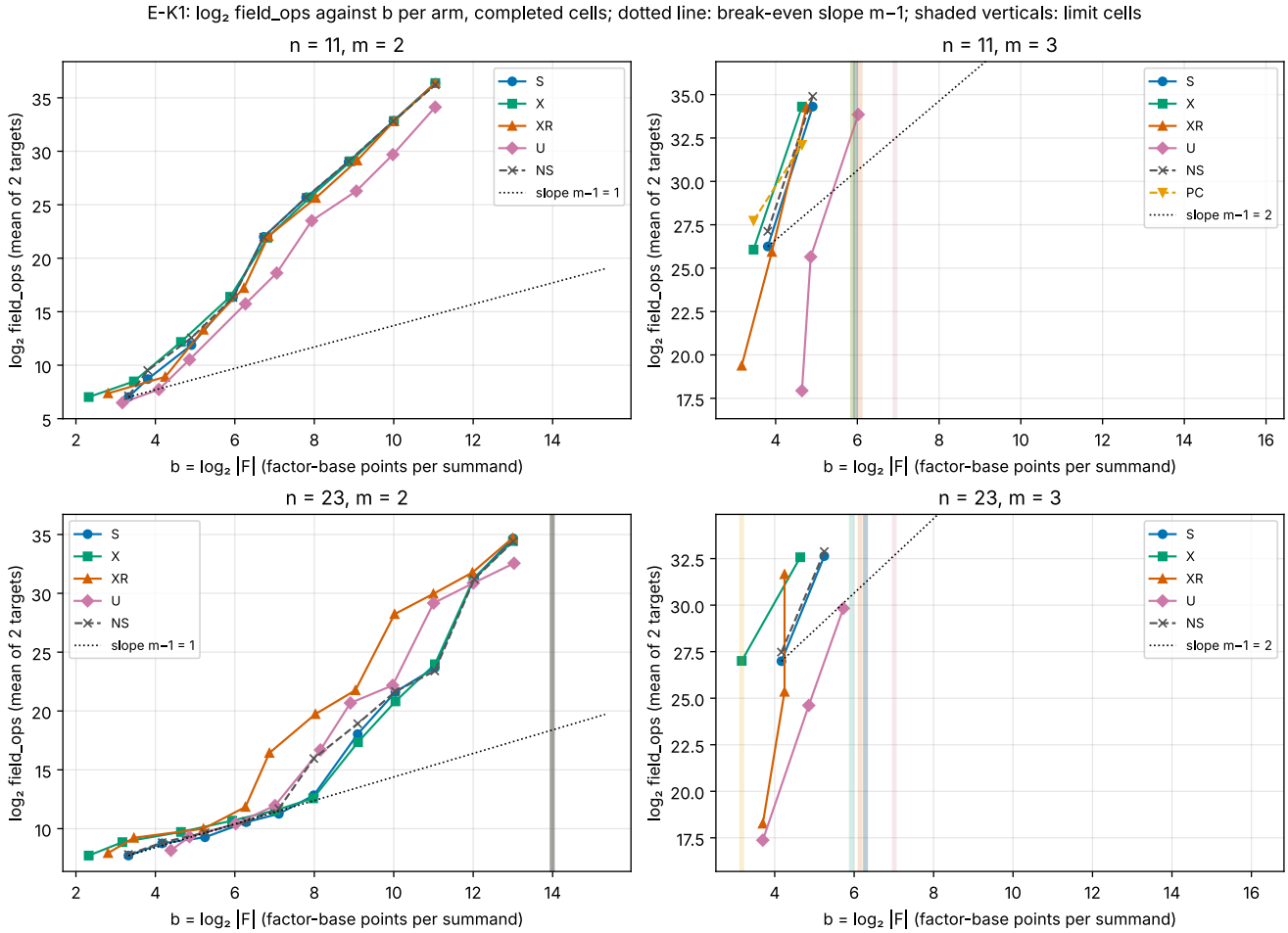


Figure 1: $\log_2$ field_ops (cell mean of 2 targets) against b for completed cells, per arm; dotted line: break-even slope m − 1 through the first S cell; shaded vertical lines: limit cells (their rows are in the records). Rows: runs/K1-*.jsonl, run-20261019T1712Z.

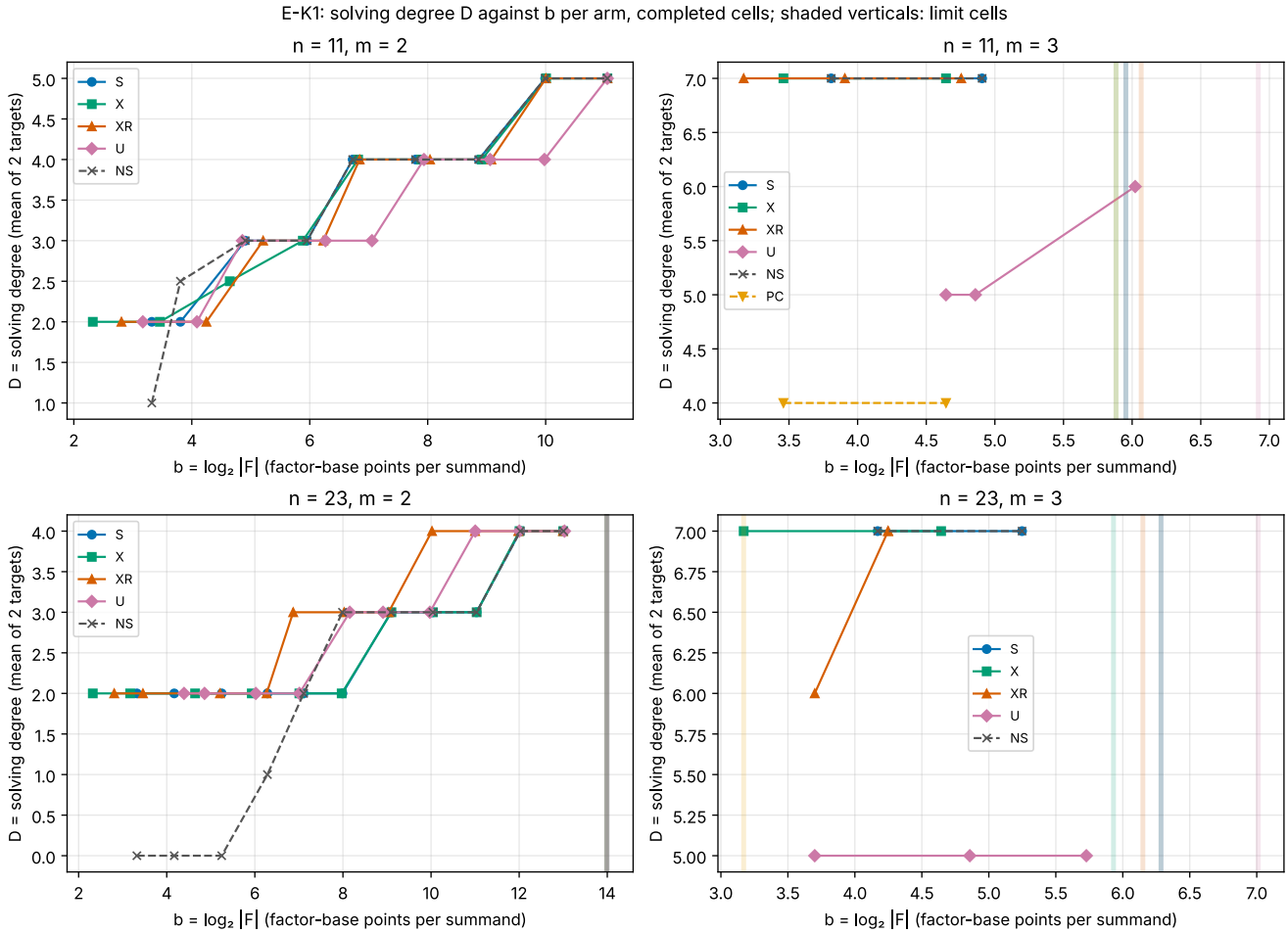


Figure 2: Solving degree D (cell mean of 2 targets) against b, same cells. At m = 3 the direct arms have input degree 6 and D = 7 at every completed cell; PC (chained, input degree 3) has D = 4; U (input degree 4) has D = 5–6.

Decision rules (analysis-amendment6.txt)

rule	n = 11, m = 2	n = 23, m = 2
R1 κ verdicts	all 5 arms unfavourable (κ 3.58–3.93)	all 5 unfavourable (κ 2.42–3.08)
R2 S vs X	9 shared cells; $\|\Delta D\| \le 1$ at every cell; κ 3.932 vs 3.581 ⟶ **differs** ($\|\Delta\kappa\| = 0.35 \ge 0.25$)	11 shared cells; $\|\Delta D\| \le 1$ at every cell; κ 2.791 vs 2.424 ⟶ **differs** ($\|\Delta\kappa\| = 0.37$)
R3 S vs XR	κ 3.932 vs 3.783 ⟶ tied; XR ≥ 2× S's field_ops in 1 of 9 shared cells, S ≥ 2× XR's in 0	κ 2.791 vs 2.849 ⟶ tied; XR ≥ 2× S's in 6 of 11, S ≥ 2× XR's in 0
R3 S vs U	κ 3.932 vs 3.642 ⟶ S higher; U ≥ 2× cheaper in 6 of 9, S in 0	κ 2.791 vs 3.077 ⟶ S lower; S ≥ 2× cheaper in 3 of 10, U in 1
R4 outcome	**no advance** (S cost-unfavourable)	**no advance** (S cost-unfavourable)
R5 null NS vs S	**mixed**: D equal in 7 of 9 cells; differs at $\ell = 3$ (NS 1.0, S 2.0) and $\ell = 4$ (NS 2.5, S 2.0)	**mixed**: D equal in 6 of 11; NS lower in 4 (ℓ = 3–6, D 0, 0, 0, 1 against 2), higher in 1 ($\ell = 8$, 3 against 2); NS ≥ 2× S's field_ops in 1 of 11
R7 Gauss vs random x-window	XR ≥ 2× X's in 1 of 9, ≤ ½ in 0	XR ≥ 2× X's in 6 of 11, ≤ ½ in 0

At m = 3 every R2–R5 comparison of S has only 2 shared completed cells and is **inconclusive** by the rule (fewer than 3); R4 is inconclusive at both n.

R6, positive control. PC1 (random x-window with direct S_4, degree 6, against the u-frame, degree 4, at m = 3, same instrument): 4 shared cells (n = 11 $\ell = 4, 5$; n = 23 $\ell = 4, 5$), XR's field_ops 255×, 390×, 251× and 132× U's, D = 7 against 5 in all four. **Detected** (≥ 2× in ≥ 2 cells), so the instrument is not blind to this difference. PC2 (descriptive): X chained against X direct, same window and planted points, 2 shared cells (n = 11 $\ell = 4, 5$): D = 4 against 7; chained field_ops 3.17× ($\ell = 4$) and 0.22× ($\ell = 5$) the direct arm's. At n = 23 PC has no completed cell ($\ell = 4$ oversize).

Trace replay. 250 of 250 completed systems, re-measured once from their seed, reproduced D, max_cols, max_rows and steps; field_ops was also equal in all 250. Every system is therefore counted in the table above.

Predictions scorecard

prediction (Amendment 6 §9)	result	reading
P1 identity of the s-line polynomial	held	exact, checked on the grid
P2 S tied with X in every (n, m)	**failed**	D within ±1 everywhere, but $\|\Delta\kappa\|$ = 0.35, 0.37 at m = 2 (m = 3 not evaluable)
P3 no arm cost-favourable	held	14 series with ≥ 3 cells, all unfavourable
P4 PC1 detects, XR dearer than U	held	4 of 4 cells, ratio 132–390×
P5 S does not advance	held	no advance at m = 2 for both n; inconclusive at m = 3
P6 null reads "structure beyond shape"	**failed**	**mixed** at m = 2 for both n; not evaluable at m = 3
P7 Gauss x-window cheaper than random in ≥ half of shared cells	**failed**	held at n = 23, m = 2 (6 of 11); 1 of 9 at n = 11; 0 of 2 at m = 3
P8 S, X, XR, NS have < 3 cells at m = 3, n = 11	**failed**	XR has 3 (random ℓ = 3 window was plantable; erratum)

Observations (descriptive; no claim beyond these rows)

- Every series with at least 3 completed cells is cost-unfavourable: 14 of 14 (10 at m = 2, κ 2.42–3.93 against 1; 4 at m = 3, κ 6.15–18.71 against 2). S's κ is 3.93 (n = 11) and 2.79 (n = 23) at m = 2.
- At m = 2, D rises from 2 to 5 (n = 11) or 4 (n = 23) over the cells of S, X, XR and U, with σ between 0.20 and 0.47 for the five arms.
- S and X have identical equations and D within ±1 at every shared cell, yet their κ differ by 0.35–0.37, more than the 0.25 margin of R3. At these sizes the spread of κ between arms with identical equations exceeds the registered margin (a possible reason: b is a count of curve points and is noisy at small ℓ; not tested here).
- Against the u-frame, S's κ is higher at n = 11 and lower at n = 23; the level at equal ℓ is mostly in U's favour at n = 11 (6 of 9 cells ≥ 2×) and mostly tied at n = 23.
- The Gauss x-window is cheaper than the random x-window at equal ℓ in the middle of the n = 23, m = 2 range: XR's field_ops are 30× X's at ℓ = 7, 170× at ℓ = 10, about 60× at ℓ = 11 and within 2× at ℓ = 12 and 13. At n = 11 the two are within 2× in 8 of 9 cells (the windows are nearly the whole field from $\ell \approx 9$).
- The random-coefficient null is not harder than S in D: at n = 23, m = 2 its D is lower at ℓ = 3–6 (D = 0 in three cells: F4 added no element after the initial echelon form of the inputs), higher at ℓ = 8 by 1 and equal at the other six cells.
- At m = 3 the direct arms (S, X, XR, NS) complete only ℓ = 4–5 (D = 7, about 2×10^{10} XORs at ℓ = 5, 19 to 26 s). ℓ = 6 hits the time budget (n = 11: 9 steps in 300 s, peak matrix 0.79 GB) or the 1 GiB matrix limit (n = 23). U reaches ℓ = 6; PC (chained) reaches ℓ = 5 at n = 11 and none at n = 23.
- At n = 23, m = 2 every arm hits the matrix limit at ℓ = 14 after 28 to 58 s.

Not shown

- Nothing about n = 131, other a, the other Gauss-window class ($\zeta \mapsto \zeta^k$), a u-frame on a Gauss-derived V, m ≥ 4, or any engine other than `groebner_basis_f4` the budgets and the 1 GiB matrix cap decide where each series stops, so the κ values are readings over the cells that finished ($\ell \le 13$).
- field_ops counts Boolean word XORs, not F_p multiplications; κ is compared with m − 1 and across arms of this amendment only.
- No pipeline is priced; no speedup, win or significance language is licensed. The m = 3 series of S, X, NS and PC are inconclusive by the rule, and XR's m = 3 κ values (9.40, 18.71) rest on an irregular b.
- The replay check shows reproducibility of the trace from the seed, not correctness of F4′s basis beyond the planted-solution check.